Dataset Doctor Analysis Report

Dataset: Test Dataset

1. Dataset Overview

Item	Value
Number of Rows	5,000
Number of Columns	32
Missing Values	0
Duplicate Rows	0
Target Column	sex_of_driver
Problem Type	Classification

2. Data Quality

Quality Measure	Result
Overall Quality Score	80 / 100
Missing Values	Excellent
Duplicate Rows	Excellent
Data Types	Excellent
Correlation	Poor
Outliers	Poor

3. Target Balance

The selected target classes are reasonably balanced.

Class	Proportion
1	0.6182
2	0.2572
3	0.1200
-1	0.0046

4. Machine Learning Results

Model	ACCURACY	PRECISION	RECALL	F1
Random Forest	0.6920	0.6447	0.6920	0.6405
Logistic Regression	0.7000	0.6773	0.7000	0.6506
Decision Tree	0.6170	0.6195	0.6170	0.6180

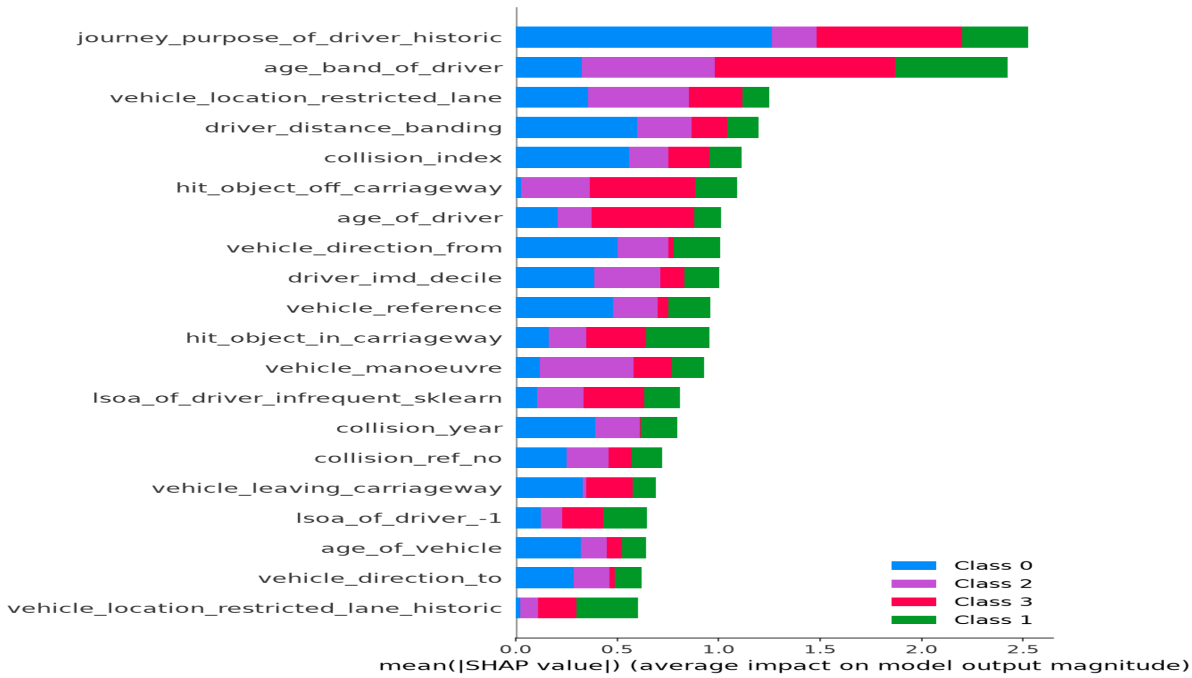
Best Model: Logistic Regression
ACCURACY: 0.7000

5. Explainable AI / SHAP

SHAP was used to explain which features had the strongest influence on the predictions made by the selected best-performing model.

Rank	Feature	Mean |SHAP Value|
1	journey_purpose_of_driver_historic	0.6309
2	age_band_of_driver	0.6061
3	vehicle_location_restricted_lane	0.3120
4	driver_distance_banding	0.2994
5	collision_index	0.2786
6	hit_object_off_carriageway	0.2725
7	age_of_driver	0.2530
8	vehicle_direction_from	0.2515
9	driver_imd_decile	0.2505
10	vehicle_reference	0.2395

SHAP Summary Visualization



6. Conclusion

The selected target column was 'sex_of_driver'. The task was identified as a classification problem. The best-performing model was Logistic Regression, with a ACCURACY score of 0.7000. The most influential features identified by SHAP were journey_purpose_of_driver_historic, age_band_of_driver, vehicle_location_restricted_lane.